ROBOTICS WITH QUARKY

TEACHER: CERTIFIED TRAINER	DURATION: 14 CLASSES (1 HOUR PER CLASS)	MODE: OFFLINE

Robotics with Quarky is an experiential, hands-on learning program designed to introduce students to the fundamentals of **robotics, sensors, motion control, and block-based programming** using **PictoBlox and the Quarky robot**.

The course blends **coding, electronics, and mechanical logic** to help learners transition from screen-based coding to real-world robotic applications. Each session focuses on **concept clarity, applied learning, and progressive skill building**, culminating in capstone projects that reinforce automation and problem-solving.

COURSE CURRICULUM

Level 1: Quarky Robotics

Session 1: Getting Started with PictoBlox

- Introduction to PictoBlox interface and workspace. Understanding how PictoBlox works with Quarky.

- Saving and managing projects
- Connecting Quarky using Bluetooth and USB. First successful device connection and interaction.

Session 2: Play Time with Sensors

- Understanding sensor-based inputs
- Mapping inputs to on-screen actions
- Learning input–output relationships
- Introduction to event-based programming. Creating the **Catch the Fruit Game**

Session 3: Meet Your AI Robot

- Revisiting PictoBlox basics and Assembling the Quarky robot
- Understanding physical robot components. Controlling robot emotions using keyboard inputs.
- Starting robot actions using space bar and stopping using alphabet keys

Session 4: Basics of Robotics

- Introduction to robot structure and anatomy
- Understanding movement principles
- Learning directional control logic. Programming robot movement in geometric shapes
- Executing square and circular motion patterns

Session 5: Getting Started with Sensors

- Introduction to different types of sensors
- Understanding how robots sense their environment
- Working with ultrasonic sensors. Measuring distance using sensor data.
- Real-time distance display and interpretation

Session 6: Detecting Obstacles

- Understanding obstacle detection concepts
- Applying conditional (if–else) logic
- Using sensors for autonomous decision-making
- Programming robot responses to obstacles. Building an obstacle avoiding robot

Session 7: Understanding Actuators

- Introduction to actuators and their role in robotics
- Difference between sensors and actuators
- Understanding motors and servo mechanisms. Servo calibration concepts
- Performing servo sweep and motion control

Session 8: Line Following Concepts

- Introduction to IR sensors
- Understanding surface and line detection

- Sensor calibration techniques. Logic behind line tracking
- Preparing sensors for line following tasks

Session 9: Line Following Robot

- Applying line following logic in automation. Using continuous loops for movement
- Real-time error correction
- Coding autonomous navigation. Building and testing a line follower robot

Session 10: Edge Detection Robot

- Understanding boundary and edge detection
- Using multiple sensors together. Implementing safety logic to prevent falls
- Programming edge-aware movement. Building an edge detection robot

Session 11: Steering and Motion Control

- Understanding differential drive systems
- Left and right motor coordination. Learning steering logic
- Autonomous directional control. Building an automatic steering robot

Session 12: Four Wheel Drive Mechanics

- Introduction to four-wheel drive systems
- Understanding power distribution
- Assembling a 4WD robot structure
- Controlling multiple motors together
- Driving and testing a four-wheel-drive robot

Session 13: Capstone – Interactive Control System

- Controlling robot movement using Quarky touch sensors
- Displaying direction of movement on LED Matrix
- Integrating sound output using Quarky speaker
- Announcing movement direction through audio
- Combining sensors, outputs, logic, and automation

Session 14: Capstone – Line Following Challenge

- Applying line following logic in a real-world environment
- Navigating a blue/green mat autonomously
- Fine-tuning sensor calibration
- Debugging and optimizing movement accuracy
- Demonstrating end-to-end robotics understanding

LEVEL 2: Quarky Robotics

Session 1: Object Tracking Robot

- Understanding AI-based object tracking concepts
- Training and testing object tracking models

- Linking visual input with continuous robot motion
- Building a real-time object tracking robot

Session 2: Pick & Place Robot – Assembly (Part 1)

- Introduction to robotic arms and gripping mechanisms
- Understanding horizontal pick and place systems
- Mechanical assembly of arm and base structure
- Preparing the system for servo control

Session 3: Pick & Place Robot – Control (Part 2)

- Understanding servo motor movement and angles
- Coding manual control for arm and gripper
- Coordinating multiple servos together
- Operating a manually controlled pick and place robot

Session 4: Building a Robo-Pet

- Introduction to AI sensing and interaction
- Implementing hand detection logic
- Mapping emotions and behaviors to inputs
- Creating an interactive robo-pet behavior system

Session 5: Self-Driving Car – Part 1

- Understanding autonomous vehicle fundamentals
- Detecting traffic signs and landmarks
- Applying object detection to navigation
- Preparing logic for decision-based movement

Session 6: Self-Driving Car – Part 2

- Implementing motion decisions based on detected signs
- Handling multiple traffic signals
- Combining AI vision with movement control
- Building a responsive self-driving robot

Session 7: Advanced Object Tracking Robot

- Revisiting face and object detection concepts
- Integrating AI tracking with robot motion
- Optimizing tracking accuracy and stability
- Building a face-tracking robot system

Session 8: Gripper Mechanism

- Understanding gripper design and functionality
- Controlling grip using servo motors
- Managing grip strength and precision
- Integrating gripper with robotic arm

Session 9: Vertical Pick & Place – Part 1
- Introduction to vertical motion systems
- Mechanical assembly of vertical lift structure
- Understanding load balance and alignment
- Preparing multi-axis movement

Session 10: Vertical Pick & Place – Part 2
- Coding vertical lift and arm movement
- Synchronizing vertical and horizontal motion
- Implementing semi-automated control
- Testing movement accuracy and repeatability

Session 11: Vertical Pick & Place – Part 3
- Full automation of pick and place cycle
- Optimizing speed and precision
- Implementing error correction logic
- Final testing of vertical pick and place system

Session 12: Weather Monitoring System
- Introduction to environmental and weather sensors
- Collecting real-time sensor data
- Interpreting environmental parameters
- Displaying weather data outputs

Session 13: Capstone – AI Robotics Application
- Identifying a real-world automation problem
- Selecting appropriate sensors and AI models
- Designing robot logic and workflow
- Building an integrated AI-based robotic solution

Session 14: Capstone – Advanced Automation Challenge
- Combining AI, sensors, and motion control
- Implementing a fully automated system
- Debugging and optimizing performance
- Demonstrating final project output

CLASS STRUCTURE (STANDARD FOR ALL SESSIONS)
- Clear explanation of session objective
- Guided hands-on coding and robot interaction
- Real-time testing and debugging
- Discussion on real-world applications